British Applied College
الكلية البريطانية التطبيقية

Using and Manipulating Data

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Learning Objectives

- Distinguish between primary and secondary data sources.
- Evaluate data reliability and accuracy.
- Understand methods for data collection and their implications.

Data Sources: Primary vs. Secondary

Primary Sources

Original sources providing direct or firsthand information about a topic or event, created by those present or involved.

- Surveys and questionnaires
- Interviews and focus groups
- Observations and experiments
- Case studies and field studies
- Original government reports and statistics
- Personal documents (diaries, letters, photos)

Secondary Sources

Sources that analyze, interpret, or summarize information from primary sources, created by someone not directly involved.

- Textbooks and reference books
- Journal articles and research papers
- Reviews and meta-analyses
- Newspapers and magazines
- Compiled government reports (third-party)
- Websites and blogs

Activity: Think-Pair-Share (10 mins)

"Identify one primary and one secondary source you used for your last assignment. Why was each useful for your research?"

The Data Reliability Checklist

- **Source Credibility**
  Determine expertise and reputation. Ensure the source is unbiased and has no conflicts of interest.

- **Consistency**
  Check if data aligns with other sources. Investigate any contradictions or discrepancies.

- **Methodology**
  Evaluate sample size, sampling method, tools, and analysis techniques for validity and reliability.

- **Reproducibility**
  Can another researcher obtain similar results by following the same methods?

- **Peer Review**
  Ensure the data has been reviewed by experts and meets scientific quality standards.

- **Transparency & Age**
  Are sources cited? Is the data still relevant (considering technological or societal changes)?

Data Collection: Observation Methods

Structured Observation

Researcher uses a specific list of behaviors or events to observe and record systematically.

Unstructured Observation

Researcher takes open-ended notes on everything they

Characteristics

- **Direct:** Provides firsthand information.
- **Objective:** Can be unbiased if recorded neutrally.
- **Detailed:** Captures rich levels of information.
- **Non-verbal:** Captures cues missed by other methods.

Implications

- **Time-consuming:** Long periods or multiple locations.
- **Researcher Bias:** Interpretation influenced by assumptions.
- **Limited Scope:** Only captures observable behaviors.
- **Ethical Concerns:** Gaining access and privacy issues.

Quick Poll

"In which scenario is observation more effective than a survey? Think about non-verbal cues."

Surveys and Questionnaires

Surveys

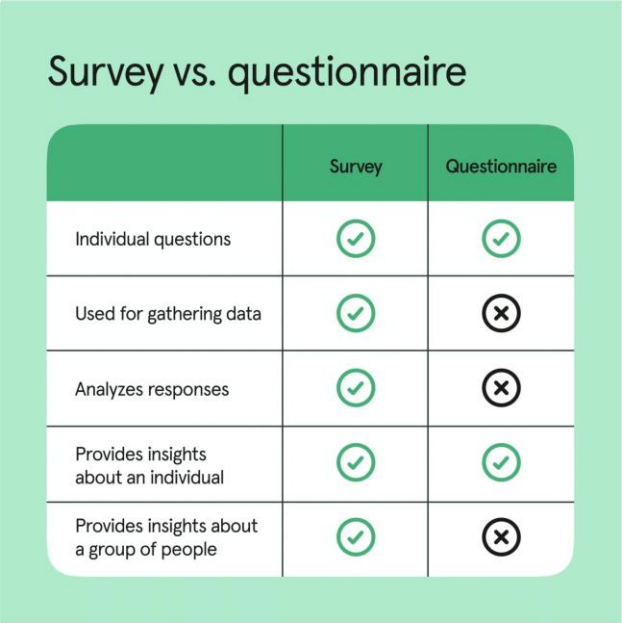
Standardized questions for large groups. Key traits: Large sample size, anonymity, and quantitative data for statistical analysis.

Questionnaires

Self-administered written surveys. Participants complete them at their own pace, ensuring standardized delivery.

Implications & Limitations

- **Limited Depth:** Closed-ended questions lack insight.
- **Response Bias:** Reluctance to answer sensitive topics.
- **Sampling Bias:** May not represent the whole population.



Scenario Question

"You are tasked with studying student study habits across the entire college. Design 3 standardized survey questions that would provide quantitative data for your analysis."

Focus Groups and Interviews

Focus Groups

Characteristics

- Interaction among participants generates new ideas.
- Flexibility for moderators to adjust discussions.
- Provides rich qualitative data on beliefs/attitudes.

Implications

- Limited generalizability due to small sample size.
- Group dynamics (dominant voices) can influence data.
- Complex analysis and limited confidentiality.

Interviews

Characteristics

- One-on-one conversations (Structured or Unstructured).
- Personal connection leads to deeper understanding.
- In-depth data for exploring complex issues.

Implications

- High risk of researcher bias in interpretation.
- Effectiveness depends heavily on interviewer skill.
- Participant honesty and comfort levels vary.

Data Accuracy and Extraction

Why Accuracy Matters

- **Research Validity:** Prevents incorrect conclusions.
- **Credibility:** Maintains researcher reputation.
- **Resource Allocation:** Ensures efficient decision-making.
- **Ethics & Compliance:** Legal requirements (e.g., Finance).

Methods of Accuracy

- **Verification:** Reviewing data for errors/inconsistencies.
- **Validation:** Comparing data to external benchmarks.
- **Automatic Systems:** Real-time error detection.
- **Quality Control:** Double-checking and calibration.

Figure 2. Data usability in relation to verification, validation, and data quality

Figure 1. Data verification and validation in relation to data quality.

Extraction & Sorting

- **Data Mining:** Statistical pattern extraction.
- **Filtering:** Selecting subsets based on criteria.
- **Pivot Tables:** Summarizing large datasets.
- **Database Queries:** Using SQL for retrieval.
- **Text Parsing:** Analyzing unstructured text (NLP).

Numerical and Data Modelling

Numerical Modelling

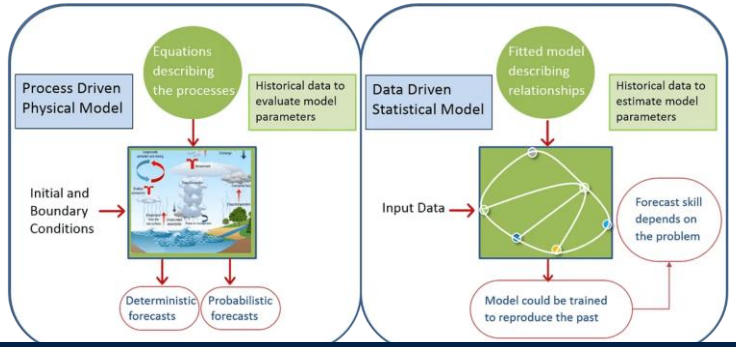
Uses mathematical models and computer algorithms to solve complex equations and simulate physical systems.

- Predicts system behavior under different conditions.
- Examples: Fluid flow in pipes, spacecraft trajectories.
- Used in engineering, physics, and finance.

Data Modelling

Involves the analysis of large datasets to identify patterns, trends, and relationships between variables.

- Uses statistical techniques and machine learning.
- Examples: Predicting consumer behavior, identifying financial risks.
- Crucial for understanding complex systems.



Discussion

"How does data modelling help companies like Netflix or Amazon predict what you want to watch or buy next?"

UI Design and Data Presentation

Presenting Results

- **Appropriate Format:** Choose tables, graphs, or charts based on data type.
- **Provide Context:** Include sources, methodology, and limitations.
- **Simplicity:** Avoid jargon; use clear and concise language.
- **Highlight Key Findings:** Focus on the most important takeaways.
- **Transparency:** Be open about assumptions and analysis constraints.

UI Characteristics

- **Ease of Use:** Simple, intuitive navigation for all expertise levels.
- **Accessibility:** Support for disabilities (high-contrast, screen readers).
- **Error Reduction:** Real-time checking and confirmation prompts.
- **Intuitiveness:** Clear actions that reduce the need for training.
- **Efficiency:** Streamlined data collection and processing workflows.

Bonus

The Challenge

"Explain the difference between **verification** and **validation** in your own words. Why are both necessary?"

The Application

"You are researching **climate change**. Name one primary and one secondary source you would use for your study."

The Debate

"Is **quantitative data** (surveys) always better than **qualitative data** (focus groups)? Discuss with your partner."